A Novel DIP-Enhanced Multi-Evidence Framework for Early Predicting Mangroves-to-Aquaculture Conversion Using Temporal Series of Multi-Sensor Satellite Imagery

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Abstract : Mangrove ecosystems are highly important coastal environments which offer protection of shorelines, carbon storage, and sustenance of biodiversity. The rapid turnover of the mangrove forests into aquaculture ponds has however been of significant concern in most tropical coastal areas. Traditional remote sensing methods tend to be based on binary classification of land cover based on few spectral indices, as the multi-stage and progressive degeneration of mangroves and development of aquaculture is difficult to measure. Also, despite the high 30 m spatial resolution of Landsat imagery and 10 m highest spectral resolution of Sentinel-2, due to mixed-pixel effects in the case of a heterogeneous coastal landscape, there is in general high false-positive detection in aquaculture ponds, bare soil, and shallow water with similar spectral characteristics. The paper offers a upgraded multi-evidence framework to overcome these restrictions in providing early warning of the conversion of mangrove to aquaculture. The framework represents the conversion lifecycle as being a process that includes five stages: intact mangrove (S1), degraded mangrove (S2), cleared land (S3), formation of pond (S4), and active aquaculture (S5). An index of composite water (CWI = MNDWI + NDWI - NDVI) has been added to enhance the process of detecting water in mixed-pixels, whereas an NDBI-based constraint is used to reduce spectral confusion of water and bare soil. Moreover, an evolutionary pipeline of post-classification DIP comprising of Discrete Wavelet Transform (DWT), Multi-Scale Retinex (MSR), Morphological Gradient edge extraction, and Guided Filtering increases visual interpretability of satellite images. Additional comparison by object-based geometric analysis derived on measures of compactness and rectangularity further differentiates artificial aquaculture ponds and natural water bodies. The framework is tested on the Coringa wildlife sanctuary area of the Godavari Delta located at India with reference data as the high-resolution basemaps. The experimental studies prove that the offered technique will cause a significant decrease in false positives in comparison with the traditional single-index thresholding algorithms and allow tracing the stages of gradual conversion and early prediction of further conversion through monitory mode. The framework offers a scaled-up early-warning mechanism to keep track of mangrove degradation and help in conserving coastal ecosystem.

Keywords: mangrove–aquaculture conversion; digital image processing; wavelet enhancement; multi-scale retinex , composite water index; multi-evidence classification; satellite remote sensing; five-stage lifecycle detection